

EPA Compliance Workshops Don't Reduce Violations

Lessons from “The Effect of Compliance Assistance on Pollution Discharges and Violations of Environmental Regulations” by Paul J. Ferraro & Jay P. Shimshack in *Journal of Policy Analysis and Management*, 2026.

Offering polluters free help to follow the rules doesn't make them more likely to follow them. New research studying hundreds of facilities that were offered free compliance workshops found the workshops didn't reduce violations.

BACKGROUND

Environmental enforcement in the U.S. has long centered on inspecting facilities, catching violations, and levying fines. **Compliance assistance** takes a different approach: regulators help facilities follow the rules in the first place, through free workshops, guidance, and technical support. The program rests on the belief that many violations stem from confusion or limited expertise rather than deliberate corner-cutting. Compliance assistance has become increasingly popular at the Environmental Protection Agency (EPA). In Ohio, facilities that attended free workshops went on to violate less often afterward. Regulators took that as a sign the approach worked and moved to expand it to other states, though **whether the workshops caused the improvement has never actually been tested.**

THE WORKSHOPS' EFFECT ON VIOLATIONS, MONTH BY MONTH

Source: Ferraro & Shimshack (2026).



Note: Only facilities that attended a workshop appear here. The month a facility was trained sets its zero line, so each dot shows whether it had more or fewer violations in a given month than it did in that training month. The estimates set aside each facility's usual violation rate, the local weather, and any swings shared across the industry; the shaded band shows the range of statistical uncertainty. If the workshops worked, the line would drop below zero after the workshop. It doesn't: in every month afterward, the band still includes "no change."

HOW THE STUDY TESTED IT

In a new study, researchers set out to test that program directly, using data on hundreds of wastewater treatment facilities regulated by the Ohio EPA. Facilities that had recently been cited for violations were invited to participate in the workshops,

which were rolled out over several months. Those that volunteered received free training on how to comply with the Clean Water Act.

Evaluating whether a policy works is hard: there is no way to see what facilities would have done without it.

Comparing violations at trained facilities before and after the workshops would credit the training with whatever else was happening at the same time. Comparing facilities that signed up against those that didn't runs into a different problem: the workshops were voluntary, so a gap between the two groups could reflect the *motivation* of the facilities that signed up rather than anything the training did.

The researchers got around this by comparing facilities trained earlier against ones that had *also* signed up but had not yet been trained. Those facilities were just as motivated and they were tracked over the same months, so any trend running through the industry showed up in both groups equally. The staggered rollout is what made those not-yet-trained facilities available as a fair benchmark. That benchmark holds only if the not-yet-trained facilities really would have followed the same path as the trained ones, which is something the researchers have to assume.

The figure asks a narrower question of the same facilities: once those industry-wide trends are stripped out, did anything change in the months around a facility's own training?

WHAT THE STUDY FOUND

The researchers found that **trained facilities were no less likely to violate than facilities that had not yet been trained.**

The same was true of pollution itself. A violation is a regulatory category, and a facility can be cited without much changing in the water. So the researchers also tracked what the plants actually discharged each month, across five pollutants. Those didn't move either. The workshops changed neither what facilities reported nor what they released into the water.

Judged only on the facilities that attended, the program appears to be a triumph: violations at those plants fell by about half in the years after the workshops. The catch is that **violations fell just as far at the facilities that had signed up but hadn't been trained yet** — the decline was already underway, and the workshops added nothing to it. A facility that spent a day in a workshop went on to break its permit about as often as it would have if it had never shown up.

Compliance assistance rests on the idea that facilities break the rules because they don't fully understand them. For these wastewater plants, that wasn't the case: they look less like confused actors waiting for help than ones weighing whether compliance is worth it. That calculation is what inspections and penalties act on, and they have a track record of reducing violations. **The study suggests that these workshops do not.**

FOR MORE INFORMATION

Ferraro, Paul J., and Jay P. Shimshack. 2026. "The Effect of Compliance Assistance on Pollution Discharges and Violations of Environmental Regulations." *Journal of Policy Analysis and Management* 45 (3): e70056. doi.org/10.1002/pam.70056.

Paul J. Ferraro is at Johns Hopkins University. Jay P. Shimshack is at the University of Virginia. Contact Jay Shimshack (jay.shimshack@virginia.edu) with questions about this research.

The Environmental Inequality Lab is a nonpartisan research group that applies a rigorous, data-driven approach to understand how our environment shapes economic opportunity and well-being.